

The inconvenient truth about flocks

Leiming Chen,^{1,*} Patrick Jentsch,^{2,†} Chiu Fan Lee,^{3,‡} Ananyo Maitra,^{4,5,§} Sriram Ramaswamy,^{6,7,¶} and John Toner^{8,9,**}

¹*School of Material Science and Physics, China University of Mining and Technology, Xuzhou Jiangsu, 221116, P. R. China*

²*Cell Biology and Biophysics Unit, European Molecular Biology Laboratory, Meyerhofstraße1, 69117 Heidelberg, Germany*

³*Department of Bioengineering, Imperial College London, South Kensington Campus, London SW7 2AZ, U.K.*

⁴*Laboratoire de Physique Théorique et Modélisation, CNRS UMR 8089, CY Cergy Paris Université, F-95032 Cergy-Pontoise Cedex, France*

⁵*Laboratoire Jean Perrin, Sorbonne Université and CNRS, F-75005, Paris, France*

⁶*Centre for Condensed Matter Theory, Department of Physics, Indian Institute of Science, Bangalore 560 012, India*

⁷*International Centre for Theoretical Sciences, Tata Institute of Fundamental Research, Bangalore 560 089 India*

⁸*Department of Physics and Institute of Theoretical Science, University of Oregon, Eugene, OR 97403¹*

⁹*Max Planck Institute for the Physics of Complex Systems, Nöthnitzer Straße 38, 01187 Dresden, Germany*

We reanalyze the hydrodynamic theory of “flocks” that is, polar ordered “dry” active fluids in two dimensions. For “Malthusian” flocks, in which birth and death cause the density to relax quickly, thereby eliminating density as a hydrodynamic variable, we are able to obtain two exact scaling laws relating the three scaling exponents characterizing the long-distance properties of these systems. We also show that it is highly plausible that such flocks display long-range order in two dimensions. In addition, we demonstrate that for “immortal” flocks, in which the number of flockers is conserved, the extra non-linearities allowed by the presence of an extra slow variable (number density) make it impossible to obtain any exact scaling relations between the exponents. We thereby demonstrate that several past published claims of exact exponents for Malthusian and immortal flocks are all incorrect.

I. INTRODUCTION

Flocks [1–4] are a spectacular and ubiquitous example of spontaneously broken rotation invariance in active matter [5–8]. Specifically, polar ordered dry (i.e., non-momentum conserving) active fluids, unlike their equilibrium counterparts—planar magnets [9, 10]—can exhibit long-range orientational order in two dimensions. [8, 11]. However, an exact analytical theory of their large-distance, long-time scaling properties [2, 4, 6] has proved elusive [11]. Such an exact calculation in two dimensions was believed possible for the “Malthusian” [12] case, in which birth and death suppress density fluctuations, making the density, a fast, non-hydrodynamic variable.

In this article, we will reanalyze the hydrodynamic theory of both Malthusian and “immortal” (that is, number-conserving) two-dimensional flocks. We focus our attention on the three scaling exponents that characterize the long-distance, long-time behaviour of these systems. These exponents are: the “dynamical” exponent z giving the scaling of time t with distance y in the direction perpendicular to the direction of mean flock motion, via the relation

$$t \propto |y|^z, \quad (\text{I.1})$$

the anisotropy exponent ζ giving the scaling of distances x along the direction of mean flock motion with distances along the direction perpendicular to the direction of mean flock motion, via the relation

$$x \propto |y|^\zeta, \quad (\text{I.2})$$

and the “roughness” exponent χ giving the scaling of the typical directional fluctuations $\delta\theta$ with distance y in the direction perpendicular to the direction of mean flock motion, via the relation

$$\delta\theta \propto |y|^\chi. \quad (\text{I.3})$$

For Malthusian flocks, we derive two exact scaling relations between these three exponents:

$$\chi - \zeta + 1 = 0, \quad (\text{I.4})$$

$$z - \zeta - 2\chi - 1 = 0. \quad (\text{I.5})$$

Contrary to claims in the literature [13, 14], it is not possible to obtain a third exact scaling relation. As a result, we cannot calculate the *correct* exponents for Malthusian flocks. However, the two scaling laws (I.4) and (I.5) are sufficient (indeed, (I.4) by itself is sufficient) to show that the predictions [2, 12] $z = 6/5$, $\zeta = 3/5$ and $\chi = -1/5$ for the dynamical, anisotropy and roughness exponents are incorrect.

We will, however, argue that Malthusian flocks very likely do display long-range order in two dimensions, and show that fluctuations in some directions of wavevector are smaller than were predicted earlier.

For immortal (i.e., number-conserving) flocks, we show that the proliferation of non-linearities associated with the conserved density field make it impossible to obtain *any* exact scaling relations between the scaling exponents. In the absence of such scaling laws, it is impossible to even make an analytic argument that long-ranged

order exists in these systems. That they do, however, is shown by numerous detailed simulations of both microscopic models whose hydrodynamic behaviour should belong to the universality class of immortal flocks (see, e.g., [15]), and continuum equations of the form discussed in this article (see, e.g., Ref. [8]). We will further show that several recent articles [13, 14, 16] claiming to calculate the exact exponents of flocks are incorrect, either because they violate certain symmetries, or because they assume symmetries and/or conservation laws that do not, in fact, hold.

Our work deals only with the effect of small fluctuations on the uniformly moving phase of the flock. The nature of the *transition* to that phase is of great interest [8, 17–21], but is not dealt with here. Likewise, we do not examine the stability of flocks to unbinding of topological defects. Thus, our work does not discuss or settle the intriguing recent suggestion [22], based on numerical simulations, that Malthusian flocks are always destroyed at long enough times by the unbinding of topological defects. There are other examples of nonequilibrium systems in which the unbinding of topological defects generically destroys an ordered state that is stable within a spin-wave theory. In two dimensions, the active smectic phase [23] is one such example.

II. EQUATIONS OF MOTION FOR MALTHUSIAN AND IMMORTAL FLOCKS

We start with the equations of motion for the local polarization $\mathbf{p}(\mathbf{x}, t)$ and number density $\rho(\mathbf{x}, t)$ of a flock, as functions of position $\mathbf{x}$ and time t :

$$\begin{aligned} \partial_t \mathbf{p} + \lambda_a (\mathbf{p} \cdot \nabla) \mathbf{p} + \lambda_b (\nabla \cdot \mathbf{p}) \mathbf{p} + \lambda_c \nabla(|\mathbf{p}|^2) = \\ U(\rho, |\mathbf{p}|) \mathbf{p} - \nabla P_1(\rho, |\mathbf{p}|) - \mathbf{p} (\mathbf{p} \cdot \nabla P_2(\rho, |\mathbf{p}|)) \\ + \mu_B \nabla(\nabla \cdot \mathbf{p}) + \mu_T \nabla^2 \mathbf{p} + \mu_A (\mathbf{p} \cdot \nabla)^2 \mathbf{p} + \xi_p. \end{aligned} \quad (\text{II.1})$$

and

$$\partial_t \rho = -\nabla \cdot (\beta \mathbf{p}) + D_1 \nabla^2 \rho + D_2 \nabla \cdot (\mathbf{p} \mathbf{p} \cdot \nabla) \rho - h(\rho, |\mathbf{p}|) + \xi_n + \nabla \cdot \xi_\rho \quad (\text{II.2})$$

In equations (II.1) and (II.2), the coefficients $\lambda_{a,b,c}$, $U(\rho, |\mathbf{p}|)$, $h(\rho, |\mathbf{p}|)$, $\mu_{B,T,A}$, $D_{1,2}$, β , and the “pressures” $P_{1,2}(\rho, |\mathbf{p}|)$ are, in general, functions of the flocker number density ρ and the magnitude $|\mathbf{p}|$ of the local polarization.

We will expand all of them to the order necessary to include all terms that are “relevant” in the sense of changing the long-distance behavior of the flock.

The ξ_p , ξ_ρ , and ξ_n terms are random, zero-mean, Gaussian white noises, with correlations:

$$\begin{aligned} \langle \xi_{p,i}(\mathbf{r}, t) \xi_{p,j}(\mathbf{r}', t') \rangle &= 2\Delta_p \delta_{ij} \delta^d(\mathbf{r} - \mathbf{r}') \delta(t - t'), \\ \langle \xi_{\rho,i}(\mathbf{r}, t) \xi_{\rho,j}(\mathbf{r}', t') \rangle &= 2\Delta_\rho \delta_{ij} \delta^d(\mathbf{r} - \mathbf{r}') \delta(t - t'), \\ \langle \xi_n(\mathbf{r}, t) \xi_n(\mathbf{r}', t') \rangle &= 2\Delta_n \delta^d(\mathbf{r} - \mathbf{r}') \delta(t - t'), \end{aligned} \quad (\text{II.3})$$

where the noise strengths $\Delta_{p,\rho,n}$ are constant parameters in the coarse-grained theory (analogous to the tempera-

ture in an equilibrium system, as they set the scale of fluctuations), and i, j label vector components.

These equations are derived purely from symmetry arguments [2, 3, 6]. However, each term they contain has a simple physical interpretation, which we now give.

We first note that (II.1) can be viewed as an equation of motion for either a polarization or a velocity, and whether a given term is allowed in the passive limit depends on which of these interpretations we choose. We opt for polarization in this article.

The $U(\rho, |\mathbf{p}|)$ term is responsible for spontaneous flock motion. Our analysis will apply to an extremely large class of U s; specifically, to all of those that satisfy $U(\rho_0, |\mathbf{p}| < p_0) > 0$, and $U(\rho_0, |\mathbf{p}| > p_0) < 0$ in the ordered phase. This last condition ensures that in the absence of fluctuations, the flock will reach a steady state with nonzero polarization magnitude $|\mathbf{p}| = p_0$, and $\rho = \rho_0$. The diffusion constants $\mu_{B,T,A}$ reflect the tendency of flockers to align with their neighbours. With the “polarization” interpretation for $\mathbf{p}$, the $P_2(\rho, |\mathbf{p}|)$ term in (II.1) can arise as the variational derivative of $\int \mathbf{p} \cdot \nabla P_3$ where P_3 is a scalar function of $\rho, |\mathbf{p}|^2$ [24]. The advective λ_a term, of course, is non-variational.

Both the term $h(\rho, |\mathbf{p}|)$, which embodies birth-death or evaporation-deposition processes, and the noise ξ_n (or, equivalently, the noise strength Δ_n) vanish in an “immortal” flock, in which, by definition, the number of flockers is conserved. The case in which both $\xi_{n,\rho}$ are non-zero is called a “Malthusian” flock [12]. Unsurprisingly, the immortal and Malthusian cases differ considerably, so we will treat them separately below.

For the Malthusian (i.e., non-number-conserving) dynamics to yield a stable steady-state density ρ_0 in the absence of fluctuations, the birth and death term $h(\rho, |\mathbf{p}|)$ must satisfy $h(\rho_0, p_0) = 0$, $(\partial_\rho h)_{\rho_0, p_0} > 0$.

All of the terms in (II.2), with the exception of $h(\rho, |\mathbf{p}|)$ and the nonconserving part ξ_n of the noise, can be written as the divergence of a current in the immortal case (in which $h(\rho, |\mathbf{p}|)$ and ξ_n vanish), as is required by number conservation. Note that this is *not* the case for the polarisation equation: many of the terms in that equation are not total divergences, since the polarisation field itself does not obey a conservation law.

We will see in the next two sections that this statement also holds for the Nambu-Goldstone mode of this system; that is, the part of the polarisation field that relaxes slowly in the ordered phase for fluctuations at very long wavelengths.

This is where reference [13] goes wrong: they claim that the non-stochastic part of the dynamics of the single angle that characterizes the Nambu-Goldstone mode in two dimensions must be a total divergence and, therefore, must vanish in the limit of small wavenumbers.

This is untrue. The *only* requirement of a Nambu-Goldstone mode - which is the small amplitude, small wavenumber fluctuation of an order parameter corresponding to a broken continuous symmetry (see, for instance, [25]) - is that it relaxes slowly in the limit of

long wavelengths with the relaxation rate vanishing in the limit of infinite wavelength in which case the mode corresponds to the global transformation of the order parameter under the broken symmetry; this by no means implies that all terms in the equation of motion for a Nambu-Goldstone mode must be expressible as total divergences. Indeed, numerous counter-examples - that is, systems whose Nambu-Goldstone modes have terms in their equations of motion that can *not* be written as total derivatives - exist: the KPZ equation [26]¹ for one, and the hydrodynamics of nematic fluids [39], crystalline solids [40], smectics [40], and discotic phases [41] for a few others. In fact, except in very special circumstances, *all* active and passive liquid-crystalline and crystalline states serve as counterexamples. We will now briefly describe how each of the systems we named is a counter-example to the claim that time derivatives of Nambu-Goldstone modes must be total divergences.

The nonlinearity in the KPZ equation

$$\partial_t H = \nu \nabla^2 H + \frac{\lambda}{2} (\nabla H)^2 + f, \quad (\text{II.4})$$

i.e., the λ term, can clearly *not* be written as a total derivative. This means that, if the assertion of [13] were correct, that λ term would be forbidden, which it is not.

Note further in this context that, if the λ term in (II.4) *was*, somehow, expressible as a total divergence, then one would be inevitably led to the conclusion that the noise strength - i.e., the correlator for the noise f in (II.4) - would be unrenormalized. In fact, it has been known [26, 42] since the KPZ equation was first proposed that the noise strength is, indeed, renormalized.

Likewise, in the immortal flocking problem we study here, the noises will also be renormalized. This has been explicitly demonstrated to one loop order for $d > 2$ Malthusian flocks [43, 44], and will certainly hold for immortal flocks as well, contrary to the claims in [13], which arise entirely from their erroneous assertion that the time derivative of any Nambu-Goldstone mode must be a total divergence.

Three of the other counter-examples we quoted involve translationally ordered systems, for which the Nambu-Goldstone modes are well known [40] to be a displacement field $\mathbf{u}(\mathbf{r}, t)$, where $\mathbf{u}$ has 3 components in a crystalline solid, 2 components in a “discotic”, and one component in a “smectic”. In all three examples, the equation of motion for the displacement field is

$$\partial_t \mathbf{u} = \mathbf{v}_{\parallel} + \dots \quad (\text{II.5})$$

where $\mathbf{v}_{\parallel}(\mathbf{r}, t)$ is the projection of the local velocity field onto the directions of broken translational symmetry, and the ellipsis denotes terms involving spatial derivatives. Here the fact that the time derivative of the Nambu-Goldstone mode is not a total divergence is even more obvious: the $\mathbf{v}_{\parallel}(\mathbf{r}, t)$ term doesn’t even involve spatial derivatives, much less total divergences. Similarly, the dynamics of the Nambu-Goldstone mode in active or driven solids [45, 46] or polar active smectics [47] are not expressible as total divergences.

Finally, for states with spontaneously broken spatial rotation symmetry, the Nambu-Goldstone mode is an angle field θ (for the simplest case of two space dimensions), and it is advected by the velocity field:

$$\partial_t \theta + \mathbf{v} \cdot \nabla \theta = \dots \quad (\text{II.6})$$

where the ellipsis contains other couplings to the velocity field as well as relaxational and stochastic terms. Importantly, $\mathbf{v} \cdot \nabla \theta$ is *not* a total derivative and a term $\theta \nabla \cdot \mathbf{v}$ whose presence could have converted this to a total derivative is forbidden because it is not consistent with rotation symmetry ($\theta \nabla \cdot \mathbf{v}$ transforms the same way as θ under rotation and a term $\propto \theta$ is clearly forbidden by rotation symmetry in the equation for $\partial_t \theta$; see also the symmetry transformation (III.5) in the next section)². A similar argument can be constructed for polar and nematic liquid crystals in higher dimensions as well³.

So the claim of [13] that the time derivative of any Nambu-Goldstone mode must be a total divergence is not only unsupported by any argument, but is in fact explicitly contradicted by some of the most well-known examples of Nambu-Goldstone modes.

The correct equations of motion for flocks are our equations (II.1) and (II.2), which lead, for immortal flocks, to an equation of motion for the Nambu-Goldstone mode that *does* contain non-total divergence terms, as we’ll see in the section (IV).

The rate of change of the Nambu-Goldstone mode in Malthusian flocks *does* prove to be writable as a total divergence. However, this is a derived result, not a consequence of imposing conservation of the Nambu-Goldstone mode at the outset. Rather, it proves to be a consequence of rotation invariance, as we’ll show in section (III), which, however, does not apply to immortal flocks.

¹ If the field H of the KPZ equation is a non-compact field, then it represents the dynamics of the Nambu-Goldstone mode of broken D -dimensional spatial translation symmetry by a $D - 1$ -dimensional membrane [25]. If it is a compact field, it can represent the Nambu-Goldstone mode of driven or active periodic systems [27, 28] or a variety of time-crystalline systems, that break parity, ranging from active XY models [29, 30] to chiral active liquid crystals [31, 32] to non-reciprocal systems [33] to driven-dissipative condensates [34–38].

² Similarly, and contrary to a claim by [16], a term $\propto \theta (\nabla \cdot \mathbf{p})$ is forbidden in the angular dynamics of polar flocks that we will discuss in the next sections.

³ In fact, even the Nambu-Goldstone mode equation that can be derived from the equilibrium model A dynamics governed by a free energy functional with a term $\propto \int \mathbf{p} \cdot \nabla P_3(\rho)$ cannot be written as a total divergence. In two dimensions, the angular dynamics has the form of (IV.6), which we will discuss in Sec. IV A, with $\lambda = g_1 = 0$. Further, even the functional derivative of the two Frank-constant free energy of a two-dimensional nematic liquid crystal [48] with respect to θ cannot be written as a total divergence. Of course, the dynamics of θ in active nematics can also not be written as a total divergence [49, 50].

We'll now describe the long-wavelength theories of our two cases: Malthusian, and immortal, starting with the simpler Malthusian case.

III. HYDRODYNAMICS OF MALTHUSIAN FLOCKS

A. Equation of motion

We are interested in the long-time, large-scale dynamics of the ordered phase of a Malthusian flock. We will begin by considering the simple limit in which the turnover rate $\tau_\rho^{-1} = (\partial_\rho h)_{\rho_0, p_0}$ is so large that the density is locked on to that value, and never changes. Later, we will argue that relaxing this constraint changes nothing in the long-wavelength limit. That is, in the language of the renormalization group, departures of the turnover rate from infinitely fast relaxation are “irrelevant”.

In the absence of noise and density fluctuations, this ordered state will simply be any one of the infinity of

uniform steady states of the equations of motion (II.1) and (II.2)

$$\mathbf{p}(\mathbf{r}, t) = p_0 \hat{\mathbf{n}}, \quad (\text{III.1})$$

where p_0 is the value of the magnitude of the polarization at which $U(\rho_0, |\mathbf{p}|)$ vanishes. Here, as a consequence of the underlying rotation invariance of our dynamics, the direction $\hat{\mathbf{n}}$ of the polarization in the steady state is completely arbitrary (which is why there is an infinity of such states).

To study fluctuations about this uniform state, we write the polarization as

$$\mathbf{p}(\mathbf{r}, t) = (p_0 + \delta p(\mathbf{r}, t))(\cos \theta(\mathbf{r}, t), \sin \theta(\mathbf{r}, t)), \quad (\text{III.2})$$

where we have chosen our coordinate system with its x -axis pointing along the direction $\hat{\mathbf{n}}$ of the spontaneously broken symmetry.

Inserting (III.2) into the equation of motion for the polarization, and expanding to linear order in δp , gives

$$(\cos \theta, \sin \theta) \partial_t \delta p + p_0 (-\sin \theta, \cos \theta) \partial_t \theta = -\alpha \delta p (\cos \theta, \sin \theta) + \dots, \quad (\text{III.3})$$

where we've defined $\alpha \equiv -p_0 \left(\frac{\partial U(\rho, |\mathbf{p}|)}{\partial |\mathbf{p}|} \right)_{\rho_0, p_0} > 0$, and the ellipsis denotes terms involving spatial derivatives of θ , and δp , as well as products of δp and spatial derivatives of θ . These terms all vanish in the limit of very small spatial gradients.

Note that the left hand side of this equation (III.3) neatly separates into orthogonal components along the unit vector $\hat{\mathbf{n}} = (\cos \theta, \sin \theta)$ along $\mathbf{p}$ and the vector $\hat{\mathbf{n}}_\perp = (-\sin \theta, \cos \theta)$ perpendicular to it. Therefore, projecting (III.3) along $\hat{\mathbf{n}}$ by taking the dot products of both sides of (III.3) with $\hat{\mathbf{n}}$ leads immediately to an equation of motion for δp :

$$\partial_t \delta p = -\alpha \delta p + \dots, \quad (\text{III.4})$$

where once again the ellipsis denotes terms that vanish in the limit of very small spatial gradients.

We see immediately from (III.4) that δp is a “fast” variable, in the sense that it relaxes with a finite lifetime $\tau = |\alpha|^{-1}$ to a value determined by the ellipsis in (III.4), similarly to the density fluctuations with the relaxation timescale τ_ρ . Note also that $\partial_t \delta p$ is negligible compared to the $|\alpha| \delta p$ term since we're considering very slow modes. Hence, that $\partial_t \delta p$ term can be dropped from (III.4). Once we do so, (III.4) becomes a simple linear equation that relates $\delta p(\mathbf{r}, t)$ to the *instantaneous* value of $\theta(\mathbf{r}, t)$ and its spatial derivatives.

A further simplification follows from recognizing that, since the ellipsis ... in (III.4) always involves spatial derivatives of θ , and we are interested in situations in which all spatial derivatives are small, because we are investigating the long-wavelength (i.e., hydrodynamic) limit, it follows that δp itself is small, and so can be dropped from the ellipsis (where it always either multiplies gradients of θ , or is differentiated itself).

Thus, the end result of this analysis of the equation of motion (III.4) for δp is that δp can be expressed entirely in terms of the local instantaneous value of θ and its spatial derivatives.

We now insert this solution for δp into the equation of motion for θ . The former can be obtained by looking at the component of the equation of motion (III.3) along the vector $\hat{\mathbf{n}}_\perp$ perpendicular to the polarization, as can be done by taking the dot products of both sides of (III.3) with $\hat{\mathbf{n}}_\perp$. Having eliminated δp as described above, this equation can only involve θ and its derivatives.

Rather than explicitly go through the algebra just described, we will instead simply note that the net result of such an analysis must be a closed set of equations of motion for θ that respect the symmetries and conservation laws of our system. In two dimensions, **the symmetries are reflection about the x -axis (that is, $y \rightarrow -y$, $\theta \rightarrow -\theta$) and spatial rotation by an arbitrary angle, i.e.,**

$$\theta(\mathbf{r}, t) \rightarrow \theta(\mathbf{r}, t) + \psi, \quad x \rightarrow x \cos \psi - y \sin \psi, \quad y \rightarrow y \cos \psi + x \sin \psi, \quad (\text{III.5})$$

where ψ is any constant rotation angle. Note that the dynamical equation for θ obtained by [14] does not possess the symmetries described here - specifically, rotational symmetry (III.5) (see, for instance, their Eq. 16) - and is therefore incorrect⁴.

The complete⁵ hydrodynamic equation of motion for θ for Malthusian flocks is

$$\partial_t \theta = \lambda \nabla \cdot (-\sin \theta, \cos \theta) + D_x \partial_x^2 \theta + D_y \partial_y^2 \theta + \xi_\theta, \quad (\text{III.6})$$

where the noise ξ_θ is Gaussian and zero-mean, with two-point correlations

$$\langle \xi_\theta(\mathbf{x}, t) \xi_\theta(\mathbf{x}', t') \rangle = 2\Delta_\theta \delta(\mathbf{x} - \mathbf{x}') \delta(t - t'). \quad (\text{III.7})$$

We have verified that systematically eliminating δp as described above recovers this form. This approach also gives relations between the parameters λ , D_x , D_y , and Δ_θ and the parameters of our original equation of motion (II.1) and (II.2). We will not bother giving those relations here, since all of the parameters of (II.1) and (II.2) are phenomenological in any case (that is, they are not calculated but, in principle, can be obtained from a fit to experiment), and one could just as well directly fit λ , D_x , D_y , and Δ_θ to experiment instead.

This equation of motion for 2D Malthusian flocks was also obtained by [13].

B. Linear theory

If we neglect the part of the λ term that is nonlinear in θ in (III.6), that equation becomes

$$\partial_t \theta = -\lambda \partial_x \theta + D_x \partial_x^2 \theta + D_y \partial_y^2 \theta + \xi_\theta. \quad (\text{III.8})$$

We can eliminate the $\lambda \partial_x \theta$ term in (III.6) by changing co-ordinates to

$$\mathbf{r}' \equiv \mathbf{r} - \lambda t \hat{\mathbf{x}} \quad (\text{III.9})$$

in a new Galilean frame moving with respect to our original frame in the direction $\hat{\mathbf{x}}$ of mean flock motion at speed λ . This changes (III.8) to

$$\partial_t \theta = D_x \partial_x^2 \theta + D_y \partial_y^2 \theta + \xi_\theta, \quad (\text{III.10})$$

where we have dropped the primes on the transformed coordinates.

We can directly read off the scaling exponents implied by the linear theory from (III.10). Since the D_x and D_y terms contain the same number of derivatives, if we

define an “anisotropy exponent” ζ by saying that characteristic distances L_x along x scale with distances L_y along y according to

$$L_x \propto L_y^\zeta, \quad (\text{III.11})$$

we clearly have

$$\zeta = \zeta_{\text{lin}} = 1; \quad (\text{III.12})$$

that is, the scaling is (according to this linear theory) isotropic.

We can also define a “dynamic” exponent z by the scaling of characteristic times τ with L_y via

$$\tau \propto L_y^z. \quad (\text{III.13})$$

Since (III.10) says that one time derivative is proportional to two y derivatives, we must have

$$z = z_{\text{lin}} = 2. \quad (\text{III.14})$$

From (III.8) and (III.7) it is straightforward to show that the spatial Fourier transform of the equal-time correlator $C^\theta(\mathbf{r}) \equiv \langle \theta(\mathbf{0}, t) \theta(\mathbf{r}, t) \rangle$ is

$$C_{\mathbf{q}}^\theta = \frac{\Delta_\theta}{q^2 (D_x \cos^2 \phi + D_y \sin^2 \phi)}, \quad (\text{III.15})$$

where ϕ is the angle between the wavevector $\mathbf{q}$ and the x -axis and $q \equiv |\mathbf{q}|$.

From this, we see that the three parameters Δ_θ , D_x , and D_y completely determine the scale of the θ -fluctuations in the linear theory. (Of course, this is hardly surprising, since they are the *only* parameters of the linear theory, except for irrelevant terms that we’ve neglected.) We will use this fact in the Dynamical Renormalization Group (DRG) of the next subsection to guide us to the most convenient choice of some otherwise arbitrary rescaling exponents.

Integrating (III.15) overall $\mathbf{q}$ gives the real-space fluctuations of θ :

$$\langle \theta^2(\mathbf{r}, t) \rangle = \frac{\Delta_\theta}{2\pi \sqrt{D_x D_y}} \ln(\Lambda L), \quad (\text{III.16})$$

where we’ve introduced an ultraviolet cutoff $q_{\text{max}} = \Lambda$ and an infrared cutoff $q_{\text{min}} = \frac{1}{L}$, where L is the system’s linear spatial extent. Typically, the “roughness” exponent χ is defined as,

$$\theta_{\text{typical}} \sim L_y^\chi \quad (\text{III.17})$$

which, for the logarithmic case of (III.16), corresponds to

$$\chi = \chi_{\text{lin}} = 0. \quad (\text{III.18})$$

We will next use the DRG to show that these linear results are radically changed by the λ nonlinearity in (III.10).

⁴ A recently published erratum [51] acknowledges that the analysis in [14] was incorrect.

⁵ We retain all relevant terms. This includes nonlinearities with one factor of ∇ ; nonlinearities with two factors of ∇ will be shown to be irrelevant.

C. Dynamical Renormalization Group (DRG) analysis

Our approach is precisely that of [42].

First we decompose the Fourier modes $\theta(\mathbf{q}, \omega)$ into a rapidly varying part $\theta^>(\mathbf{q}, \omega)$ and a slowly varying part $\theta^<(\mathbf{q}, \omega)$ in the equation of motion (III.6). The rapidly varying part is supported in the cylindrical momentum shell $-\infty < q_x < \infty$, $\Lambda b^{-1} < |q_y| < \Lambda$, where Λ is the ultraviolet cutoff, and b is an arbitrary rescaling factor that we will ultimately take to be $b = 1 + d\ell$, with $d\ell$ infinitesimal. The slowly varying part is supported in $-\infty < q_x < \infty$, $0 < |q_y| < b^{-1}\Lambda$. We separate the noise ξ_θ in exactly the same way.

The DRG procedure then consists of two steps. In step 1, we eliminate $\theta^>(\mathbf{q}, \omega)$ from (III.6). We do this by solving iteratively for $\theta^<(\mathbf{q}, \omega)$. This solution is a perturbative expansion in $\theta^>(\mathbf{q}, \omega)$, as well as the fast components $\xi_\theta^>$ of the noise. As usual, the perturbation theory can be represented by Feynman graphs. We substitute these solutions into (III.6) and average over the short wavelength components $\xi_\theta^>(\mathbf{q}, \omega)$ of the noise ξ_θ , which gives a closed EOM for $\theta^<(\mathbf{q}, \omega)$.

In step 2, we rescale the *real space* field $\theta^<(\mathbf{r}, t)$, time t , and coordinates x and y as follows:

$$y \rightarrow by, \quad t \rightarrow b^z t, \quad x \rightarrow b^\chi x, \quad \theta \rightarrow b^\zeta \theta. \quad (\text{III.19})$$

At this point, the choice of the rescaling exponents z , ζ , and χ is arbitrary. However, standard DRG arguments [42] show that the values of these exponents that produce stable fixed points of the RG process are also the values of ζ , z , and χ defined by the scaling laws (III.11), (III.13), and (III.17).

After these two RG steps, we reorganize the resultant EOMs so that they have the same form as (III.6), but with various coefficients renormalized. Note that the renormalized equation *must* have the same symmetries as the original (III.6).

This reorganization amounts to multiplying the EOM by a power of b chosen to restore the coefficient of $\partial_t \theta$ to unity.

The RG now proceeds by iterating this process.

It is useful in performing the DRG to expand the equation of motion (III.6) in powers of θ . This is easily seen to lead, after the boost (III.9), to

$$\partial_t \theta = \sum_{n=1}^{\infty} a_n \theta^{2n} \partial_x \theta + \sum_{n=0}^{\infty} c_n \theta^{2n+1} \partial_y \theta + D_x \partial_x^2 \theta + D_y \partial_y^2 \theta + \xi_\theta, \quad (\text{III.20})$$

where the expansion coefficients before we begin renormalizing (i.e., on DRG step $m = 0$, where m counts the number of iterations of the DRG) are given by

$$a_n(m=0) = -\frac{(-1)^n \lambda}{(2n)!}, \quad c_n(m=0) = -\frac{(-1)^n \lambda}{(2n+1)!}. \quad (\text{III.21})$$

These expressions will not continue to hold as we iterate the DRG, although it is important to note that, because

of rotation invariance, it is only the rescaling step of the DRG that changes the ratios $a_n/a_{n'}$, $c_n/c_{n'}$, and, most importantly, $c_n/a_{n'}$. That is, the “intermediate” coefficients a_n^I and c_n^I obtained *after* averaging out the short wavelength modes but *before* rescaling must satisfy the equation $a_n^I/c_n^I = a_n/c_n$. Note that there would have been an $n = 0$ term in the first (a_n) sum had we not done the boost (III.9), which eliminated it.

The recursion relations between the parameters of (III.20) on successive steps of the RG are

$$a'_n = b^{2n\chi+z-\zeta} (a_n + \text{graphs}), \quad (\text{III.22})$$

$$c'_n = b^{(2n+1)\chi+z-1} (c_n + \text{graphs}), \quad (\text{III.23})$$

$$D'_x = b^{z-2\zeta} (D_x + \text{graphs}), \quad (\text{III.24})$$

$$D'_y = b^{z-2} (D_y + \text{graphs}), \quad (\text{III.25})$$

$$\Delta'_\theta = b^{z-\zeta-2\chi-1} (\Delta_\theta + \text{graphs}), \quad (\text{III.26})$$

where the primes denote parameters after the RG step in question, and “graphs” denote the renormalizations arising from averaging over the short-wavelength degrees of freedom.

Even without evaluating the graphical corrections in these expressions (III.22)-(III.26), we can use them to determine which of the nonlinearities a_n and c_n are “relevant”, in the RG sense of changing the behaviour of the system from that predicted by the linear theory described earlier. We can do so by first noting that, in the absence of those nonlinearities, the graphical corrections must vanish, since the nonlinearities are the only terms in the equations of motion that couple modes at different wavevectors, and, hence, the only couplings between the “fast” and “slow” modes. Hence, if we choose the rescaling exponents z , ζ , and χ in (III.24)-(III.26) to keep the linear terms $D_{x,y}$ and the noise strength Δ_θ fixed (so that the scale of the fluctuations remains the same upon renormalization), then we can assess the relevance or irrelevance of the nonlinearities a_n and c_n when those nonlinearities are initially small (so that the graphical corrections can be ignored in the early RG steps) by asking simply whether or not they grow upon renormalization if they start out being small.

Ignoring the graphical corrections, we see that equations (III.24), (III.25), and (III.26) imply that, to keep $D_{x,y}$ and the noise strength Δ_θ fixed, we must choose the rescaling exponents z , ζ , and χ so that the following three conditions are satisfied:

$$z - 2 = 0, \quad z - 2\zeta = 0, \quad z - \zeta - 2\chi - 1 = 0, \quad (\text{III.27})$$

whose solutions are trivially found to be

$$z = 2, \quad \zeta = 1, \quad \chi = 0, \quad (\text{III.28})$$

which the alert reader will recognize as the exponents we found earlier by inspecting the correlation function predicted by the linear theory.

Using these exponents in (III.22) and (III.23), we get

$$a'_n = b(a_n + \text{graphs}), \quad (\text{III.29})$$

$$c'_n = b(c_n + \text{graphs}). \quad (\text{III.30})$$

We see from this immediately that *all* of the nonlinearities a_n and c_n are *relevant* (that is, grow upon renormalization) if they are initially small (so that the “graphs” can be neglected). **This proves that the linear results (including (III.28)), are *never* valid in a two dimensional Malthusian flock, except in the very special case in which $\lambda = 0$, which will never occur in practice.**

The long wavelength physics will therefore always differ from that of the linear theory, and must be controlled by an RG fixed point at which at least one of the nonlinearities a_n and c_n are nonzero.

We’ll now argue that only two of the infinity of nonlinearities in (III.20) actually need to be considered; namely, a_1 and c_1 . To see this, we begin by arguing that the graphical corrections to Δ_θ are 0. This follows from the fact that all of the nonlinear terms in (III.20) are total divergences. **Hence, they can only lead to renormalizations of the noise that are total divergences, while the noise ξ_θ is not. Hence, the contribution of those nonlinearities to the Fourier transformed noise correlation $\langle |\xi(\mathbf{q}, \omega)|^2 \rangle$ must be proportional to at least two powers of $\mathbf{q}$. Since Δ_θ is independent of $\mathbf{q}$, it therefore cannot be renormalized by any of the nonlinear terms. Hence the “graphs” in equation (III.26) must vanish.**

Therefore, to obtain a fixed point, equation (III.26) implies that we must have

$$z - \zeta - 2\chi - 1 = 0 \quad (\text{III.31})$$

even at any *nonlinear* fixed point. A $4 - \epsilon$ expansion for Malthusian flocks [43, 44] finds that, near four spatial dimensions, the correction to the diffusion constants $D_{x,y}$ are all positive. It seems plausible to assume that this remains true all the way down to $d = 2$.

This implies that, to keep D_y fixed, we must have

$$z - 2 < 0, \quad (\text{III.32})$$

while to keep D_x fixed, we must have

$$z - 2\zeta < 0. \quad (\text{III.33})$$

This last condition clearly implies

$$\frac{z}{2} < \zeta, \quad (\text{III.34})$$

while (III.32) implies

$$z < 2. \quad (\text{III.35})$$

Together with (III.31), (III.34) implies

$$\chi < \frac{z}{4} - \frac{1}{2}. \quad (\text{III.36})$$

Using (III.35) in this implies

$$\chi < 0. \quad (\text{III.37})$$

This is an extremely important result, since it implies that the long-range ordered state survives fluctuations.

That is, it says that the characteristic fluctuations of $\theta(\mathbf{r}, t)$ in real space, whose L_y -dependent part scales like L_y^χ , remain finite as $L_y \rightarrow \infty$.

For the purposes of identifying the relevant nonlinearities in (III.20), the negativity of χ implies, as can be immediately seen from (III.22) and (III.23), that the most relevant terms of the set a_n, c_n in (III.20) are those with the smallest value of n since the graphical corrections are the same for all n as required by rotation invariance; that is, $n = 1$ for a_n and $n = 0$ for c_n (we can’t eliminate a_1 using this argument even though its recursion relation has a higher power of χ than the recursion relation for c_0 because we don’t know the value that ζ will assume at the fixed point, where we have argued $\chi < 0$).

Therefore, the minimal theory that predicts the large-distance, long-time behaviour of such flocks is of the form

$$\partial_t \theta = \lambda_1 \partial_y \left(\frac{\theta^2}{2} \right) + \lambda_2 \partial_x \left(\frac{\theta^3}{6} \right) + D_x \partial_x^2 \theta + D_y \partial_y^2 \theta + \xi_\theta, \quad (\text{III.38})$$

with, before renormalization, $\lambda_1 = -\lambda = -\lambda_2$.

This is the model we will now analyse using the DRG.

We begin by rewriting the recursion relations in the form of differential equations. This very standard approach proceeds as follows: as already mentioned, we choose $b = 1 + d\ell$ with $d\ell$ differential. Instead of keeping track of the number n of iterations of the renormalization group, we introduce a “renormalization group time” ℓ defined as $\ell \equiv nd\ell$. This renormalization group “time” should *not* be confused with the actual time t in our problem; the two are completely different! And, instead of speaking of the values of the parameters after n iterations, we think of them as continuous functions of the continuous variable ℓ .

Taking the difference between successive values of the parameters and dividing by $d\ell$ leads to differential recursion relations for the parameters, which, based entirely on equations (III.22)-(III.26), and our earlier argument that there are no graphical corrections to Δ_θ , can be written

$$\frac{dD_x}{d\ell} = \left[z - 2\zeta + f_{D_x}(\lambda_1, \lambda_2, D_x, D_y, \Delta_\theta) \right] D_x, \quad (\text{III.39a})$$

$$\frac{dD_y}{d\ell} = \left[z - 2 + f_{D_y}(\lambda_1, \lambda_2, D_x, D_y, \Delta_\theta) \right] D_y, \quad (\text{III.39b})$$

$$\frac{d\Delta_\theta}{d\ell} = \left(z - \zeta - 2\chi - 1 \right) \Delta_\theta, \quad (\text{III.39c})$$

$$\frac{d\lambda_1}{d\ell} = \left[\chi + z - 1 + f_\lambda(\lambda_1, \lambda_2, D_x, D_y, \Delta_\theta) \right] \lambda_1, \quad (\text{III.39d})$$

$$\frac{d\lambda_2}{d\ell} = \left[2\chi + z - \zeta + f_\lambda(\lambda_1, \lambda_2, D_x, D_y, \Delta_\theta) \right] \lambda_2, \quad (\text{III.39e})$$

where the f_{D_x} etc denote the “graphical corrections”.

Two features of these recursion relations require some explanation:

1) the recursion relation (III.39c) for Δ_θ is *exact*, since, as shown earlier, there are no graphical corrections to Δ_θ . As we discussed earlier, this implies the exact scaling relation (III.31).

2) The graphical corrections f_λ to λ_1 and λ_2 are equal. This follows from the underlying rotation invariance of the dynamics. Specifically, the original form of the non-linearity $\nabla \cdot (-\sin \theta, \cos \theta)$ is fixed by rotation invariance and implies a relation amongst the coefficients of the powers of θ upon expanding the trigonometric functions. That is, if we write $\lambda_1 \theta \partial_y \theta \equiv -\lambda \theta \partial_y \theta$, $\lambda_2 \theta^2 \partial_x \theta \equiv (\lambda \theta^2 / 2) \partial_x \theta$, the ratio between λ_1 and λ_2 must be preserved under the perturbative step of the RG. Of course, these ratios can and will trivially change on the rescaling step.

Now consider in light of this observation what happens when we perform the first part of this RG step, in which we integrate out large-wavenumber modes. The “intermediate” values of $\lambda_i^I(\ell + d\ell)$, where $i = 1, 2$, after the integration part of the RG step that takes us from RG “time” ℓ to $\ell + d\ell$, but before the rescaling part of this step, can be written in complete generality

$$\lambda_i^I(\ell + d\ell) = \lambda_i(\ell + d\ell)[1 + f_i(\lambda_1, \lambda_2, D_x, D_y, \Delta_\theta)d\ell]. \quad (\text{III.40})$$

But since the ratio of λ_1 to λ_2 is to be preserved on this step, we must have

$$f_1(\lambda_1, \lambda_2, D_x, D_y, \Delta_\theta) = f_2(\lambda_1, \lambda_2, D_x, D_y, \Delta_\theta). \quad (\text{III.41})$$

Performing the rescaling part of the RG and constructing the differential recursion relations as described above then immediately implies (III.39d) and (III.39e), with f_λ the common value of f_1 and f_2 .

We can argue that f_λ must vanish if λ_2 vanishes. To show this, we begin by noting that when $\lambda_2 = 0$, Eq. (III.38) has a “pseudo-Galilean invariance” under the transformation $\theta \rightarrow \theta + \theta_0$, $y \rightarrow y + \lambda_1 \theta_0 t$, with θ_0 being an arbitrary constant. This invariance must continue to hold, *with the same value of λ_1* , under renormalisation [3, 11, 12]. This implies that f_λ must vanish when $\lambda_2 = 0$. Analyticity in both λ ’s then implies that f_λ can be no larger than order λ_2 for small λ_2 .

Now let us consider the possible types of fixed points of these recursion relations.

Since we have already shown that a fixed point with all of the nonlinearities equal to zero is always unstable under renormalization, only three possibilities for the values $(\lambda_1^*, \lambda_2^*)$ of $\lambda_{1,2}$ at a stable fixed point remain:

- i) $\lambda_1^* \neq 0$, $\lambda_2^* = 0$.
- ii) $\lambda_1^* = 0$, $\lambda_2^* \neq 0$.
- iii) $\lambda_1^* \neq 0$, $\lambda_2^* \neq 0$.

We will now show that only possibility (iii) is, in fact, possible. We’ll show this by demonstrating that fixed points of the type (i) and (ii) are unstable under renormalization.

Consider the first possibility (i), which is the type of fixed point investigated by Toner [12] (who, alas, failed to recognize that it was unstable). At such a fixed point, the equation of motion (III.38) is identical to a model for a completely unrelated physical system studied by Hwa and Kardar [52].⁶ Any such fixed point must be unstable against λ_2 . To see this, note that, at such a fixed point, if we linearize the recursion relation (III.39e) for λ_2 , the f_λ term drops out. This is because, as noted above, f_λ is at least of order λ_2 at such a fixed point. Hence, its contribution to the recursion relation for λ_2 at a fixed point of the type (i) is of $O(\lambda_2^2)$ at most, and hence vanishes upon linearization.

This leaves the recursion relation for λ_2 as

$$\frac{d\lambda_2}{d\ell} = (2\chi + z - \zeta) \lambda_2. \quad (\text{III.42})$$

At a fixed point of the type (i), it is possible to actually determine the exponents exactly. One can do this by noting that, as we argued earlier, f_λ vanishes at such a fixed point due to pseudo-Galilean invariance, since $\lambda_2 = 0$ there. Equation (III.39d) therefore implies that, to keep λ_1 fixed and non-zero (which, by assumption, it is at this fixed point), we must have

$$\chi + z - 1 = 0. \quad (\text{III.43})$$

In addition, when λ_2 is zero, as we are assuming for type (i) fixed points, D_x cannot renormalize, since the only nonlinearity is a total y -derivative, which cannot generate a term like the D_x in (III.20), which is a total x derivative. Hence, the graphical correction f_{D_x} in the recursion relation (III.39a) for D_x must vanish at the fixed point (i). Therefore, to keep D_x fixed, (III.39a) implies

$$z = 2\zeta. \quad (\text{III.44})$$

Note that we now have three linearly independent equations ((III.43), (III.44), and (III.31)) for the three exponents χ , ζ , and z . Solving these gives

$$\chi = -\frac{1}{5}, \quad \zeta = \frac{3}{5}, \quad z = \frac{6}{5}, \quad (\text{III.45})$$

which are the exponents obtained by Toner [12].

Unfortunately, as can be immediately seen from the linearized recursion relation (III.42) for λ_2 at a fixed point of the type (i), these exponents imply that such a fixed point is unstable against λ_2 . Indeed, plugging the exponents (III.45) into (III.42) gives

$$\frac{d\lambda_2}{d\ell} = \frac{1}{5} \lambda_2, \quad (\text{III.46})$$

⁶ Hwa and Kardar’s model was *not* rotation invariant to start with and, therefore, the considerations that forced us to retain λ_2 didn’t apply.

which clearly shows that this fixed point is unstable.

We note in passing that in an “easy-plane” Malthusian flock in three dimensions [53], which has a very similar structure to the two-dimensional Malthusian flock we study here, an analysis like that just presented implies that λ_2 is actually *irrelevant* at a fixed point of type (i), which is therefore stable. Exact exponents *can*, therefore, be found for that problem. But there’s no such luck here.

We’ll now show that any fixed point of the type (ii) is unstable under renormalization.

We start by noting that, in case ii, since λ_1 is zero, D_y cannot renormalize, since the only nonlinearity is a total x -derivative, which cannot generate a term like the D_y term in (III.38), which is a total x derivative. Hence, the graphical correction f_{D_y} in the recursion relation (III.39b) for D_y must vanish at the fixed point (ii). Therefore, to keep D_y fixed, (III.39b) implies

$$z = 2. \quad (\text{III.47})$$

Furthermore, the exact recursion relation for the noise strength Δ_θ implies that at a fixed point, we must have

$$z - 2\chi - 1 - \zeta = 0. \quad (\text{III.48})$$

Using our earlier result (III.47) that $z = 2$ in this condition (III.48), and solving for the roughness exponent χ gives

$$\chi = \frac{1 - \zeta}{2}. \quad (\text{III.49})$$

Since, as we argued earlier, and as vast numerical evidence suggests, $\chi < 0$, (III.49) implies that $\zeta > 1$ for a fixed point of type ii.

Since λ_2 goes to a non-zero value at this fixed point, we can say, from its recursion relation equation (III.39e), that the fixed point value f_λ^* of f_λ at such a fixed point must obey

$$2\chi + z - \zeta + f_\lambda = 0, \quad (\text{III.50})$$

which, taken together with our result $z = 2$ at fixed points of type ii, implies

$$3 - 2\zeta = -f_\lambda^*. \quad (\text{III.51})$$

Now, using our results (III.47), (III.49), and (III.51) in the recursion relation (III.39d) for λ_1 gives

$$\frac{d\lambda_1}{d\ell} = [\chi + z - 1 + f_\lambda^*]\lambda_1 = \left[\frac{3(\zeta - 1)}{2}\right]\lambda_1. \quad (\text{III.52})$$

Since, as we established earlier, the negativity of χ requires $\zeta > 1$, $3(\zeta - 1)/2$ is positive, and therefore, any fixed point of type ii is unstable to any non-zero λ_1 .

Thus, we are left with possibility (iii); that is, fixed points at which both λ_1 and λ_2 are non-zero. From equations (III.39d) and (III.39e), we see that, at such a fixed point, we must have

$$\chi + z - 1 + f_\lambda(\lambda_1^*, \lambda_2^*, D_x^*, D_y^*, \Delta_\theta^*) = 0, \quad (\text{III.53})$$

$$2\chi + z - \zeta + f_\lambda(\lambda_1^*, \lambda_2^*, D_x^*, D_y^*, \Delta_\theta^*) = 0 \quad (\text{III.54})$$

Subtracting (III.53) from (III.54) gives us another scaling relation between the exponents χ and ζ :

$$\chi - \zeta + 1 = 0. \quad (\text{III.55})$$

This, taken together with the relation (III.31) (which we repeat here for the reader’s convenience):

$$z - \zeta - 2\chi - 1 = 0 \quad (\text{III.56})$$

gives us two exact scaling laws relating the three universal exponents χ , ζ , and z . These *quantitative* predictions could be tested against experiments.

A further, more qualitative, experimental prediction concerns the form of the static structure factor. Using the standard trajectory integral matching technique [4, 54], we can write the static structure factor of angular fluctuations in a form that looks exactly like the one from the linear theory, but with wavevector-dependent quantities $D_x(\mathbf{q})$ and $D_y(\mathbf{q})$ (but a noise strength Δ_θ that is wavevector-independent at small wavenumbers, since it is not renormalised):

$$C_{\mathbf{q}}^\theta = \frac{\Delta_\theta}{D_x(\mathbf{q})q_x^2 + D_y(\mathbf{q})q_y^2}. \quad (\text{III.57})$$

Here, the result of integrating out the small wavelength modes is contained in the “renormalised” $D_x(\mathbf{q})$ and $D_y(\mathbf{q})$. We cannot calculate these renormalised quantities, but we expect *both* of them to diverge at small wavenumbers (since nothing protects either of them from renormalisation and, as argued earlier, both of them should increase under renormalisation). This implies that $\langle |\theta(\mathbf{q}, t)|^2 \rangle$ should diverge at small wave numbers with a power that is smaller than 2 for all wavevector directions. This is, of course, consistent with our earlier result that the roughness exponent $\chi < 0$.

Note also that this divergence is much weaker, for $\mathbf{q}$ along the direction x of mean flock motion, than the $1/q^2$ divergence in that direction found by [3, 11], as pointed out in the introduction.

If we were able to obtain an independent third exact scaling relation, we could go further and predict the values of all three of these exponents. This can be done, for example, for 3D easy plane Malthusian flocks [53], for which the irrelevance of the λ_2 vertex implies that D_x gets no graphical corrections, which provides one with a third exact scaling relation.

Unfortunately, we have been unable to obtain such a third exact scaling relation. Solon and Chate [13] claim that such a relation follows from a “pseudo-Galilean invariance” of the equation of motion (III.6). However, the equation actually does not have the symmetry they claim, as we will now show.

Their argument begins by writing the equation of motion (III.38) before renormalization in the form

$$\partial_t \theta = -\lambda \left(\theta \partial_y \theta - \frac{1}{2} \theta^2 \partial_x \theta \right) + D_x \partial_x^2 \theta + D_y \partial_y^2 \theta + \xi_\theta. \quad (\text{III.58})$$

They then assert that this equation has an invariance under a transformation in which a constant shift of θ is compensated by a moving shift of the spacetime coordinates. That is, if $\theta(x, y, t)$ is a solution of (III.58), there is a family of solutions $\theta'(x', y', t') = \theta(x + x_c, y + y_c, t + t_c) + \theta_0$, for non-constant x_c, y_c, t_c . Replacing θ by $\theta + \theta_0$ in (III.58), we get

$$\begin{aligned} \partial_t \theta &= -\lambda \left[(\theta + \theta_0) \partial_y \theta - \frac{1}{2} (\theta + \theta_0)^2 \partial_x \theta \right] \\ &\quad + D_x \partial_x^2 \theta + D_y \partial_y^2 \theta + \xi_\theta. \end{aligned} \quad (\text{III.59})$$

From the form of (III.59), at first sight, it seems that taking

$$y_c = -\lambda \theta_0 t, \quad x_c = \lambda \left[\theta_0 \int_0^t \theta d\bar{t} + \frac{\theta_0^2 t}{2} \right], \quad t_c = 0 \quad (\text{III.60})$$

then recovers our original equation, which suggests that the equation of motion (III.38) does have a pseudo-Galilean invariance. Specifically, the time-derivative becomes

$$\partial_t \theta \equiv \partial_{t'} \theta - \lambda \left[\theta_0 \partial_{y'} - \left(\theta_0 \theta + \frac{\theta_0^2}{2} \right) \partial_{x'} \right] \theta, \quad (\text{III.61})$$

apparently cancelling the θ_0 -dependent parts on the R.H.S. of (III.59).

However, because their transformation of the coordinates involves the field θ , which is a function of space and time, and not simply the parameter θ_0 and time, $\partial_{x'} \theta \neq \partial_x \theta$, and $\partial_{y'} \theta \neq \partial_y \theta$, in contrast to the usual Galilean transformations [12]. Instead,

$$\partial_x \theta = \left(1 + \lambda \theta_0 \partial_x \int_0^t \theta d\bar{t} \right) \partial_{x'} \theta \quad (\text{III.62})$$

and

$$\partial_y \theta = \partial_{y'} \theta + \left(\lambda \theta_0 \partial_y \int_0^t \theta d\bar{t} \right) \partial_{x'} \theta. \quad (\text{III.63})$$

This implies that the terms on the R.H.S. of (III.59) do not remain unchanged upon the transformation of coordinates to (x', y', t') ; instead they transform in a complicated, field-dependent way.

As a result, it is incorrect to argue, as [13] do, that the equation possesses the claimed pseudo-Galilean invariance which protects $\lambda_{1,2}$ from graphical renormalization. In fact, we expect that $\lambda_{1,2}$ *will* get graphical renormalizations (i.e., that f_λ in equations (III.39d) and (III.39e) is non-zero. As a result, we cannot obtain a third exact scaling relation, and, hence, have no simple way to determine the exact values of the exponents χ , ζ , and z for two-dimensional Malthusian flocks.

Nor have we been able to find any alternative approach to determining these exponents, even approximately. The obvious approach of actually calculating the graphical corrections can, unfortunately, only be done perturbatively in the nonlinearities. This has been done for our original model equations (II.1) and (II.2) with the birth and death term $h(\rho) \neq 0$ in a $d = 4 - \epsilon$ -expansion by [43, 44], who obtained exponents that are quite trustworthy in three dimensions; it was not necessary to take the λ_2 nonlinearity into account for this calculation since it is *irrelevant* near $d = 4$ i.e., the fixed point for dimensions near 4 is of type (i)⁷. However, the perturbation theory, as usual in ϵ expansions, gets less reliable as one goes to lower spatial dimensions, and certainly is completely untrustworthy by the time one reaches $d = 2$. So we unfortunately can say nothing quantitative about the values of the exponents for 2d Malthusian flocks, other than that they obey the two exact scaling relations (III.55) and (III.56).

D. Justification for ignoring density fluctuations in the Malthusian flock

We now show that density fluctuations in Malthusian flocks can be ignored, as we have done in this section. Retaining only the most relevant terms, the dynamical equation for density fluctuations $\delta\rho$ about its steady state value ρ_0 is given by

$$\partial_t \delta\rho = -\beta(\rho_0) \partial_y \theta - h'(\rho_0) \delta\rho + \xi_n. \quad (\text{III.64})$$

From this, we can solve for $\delta\rho$:

$$\delta\rho \approx \frac{\xi_n - \beta(\rho_0) \partial_y \theta}{h'(\rho_0)}, \quad (\text{III.65})$$

We now examine how density fluctuations affect the θ equation. To do this, we have to obtain the terms involving the density field in that equation. We do this in detail in Sec. IV A. Anticipating that discussion, we see that we have (i) a linear term $\propto c_s \partial_y \delta\rho$, (ii) nonlinear terms of the form $\partial_y (\delta\rho)^2$ and (iii) nonlinear terms that involve some power(s) of θ , one gradient and $\delta\rho$. Representative examples are $\theta \partial_x \delta\rho$, $\delta\rho \partial_x \theta$, $\theta^2 \partial_y \delta\rho$ etc. Using Eq. (III.65), we find that the only effect of the linear term $c_s \partial_y \delta\rho$ is to change the value of D_y by a *finite* amount. However, since D_y was a phenomenological coefficient, to begin with, this doesn't change our previous conclusions.

Ignoring graphical corrections, the coefficients of nonlinearities of type (ii) above (i.e., those $\propto \partial_y (\delta\rho)^2 \sim \partial_y (\partial_y \theta)^2$) scale as $b^{z-3+\chi}$, rendering them irrelevant.

⁷ Using the dimension-dependent versions of the linear exponents χ_{lin} , ζ_{lin} , z_{lin} in Eq. (III.39e) with $f_\lambda = 0$, we find that λ_2 becomes relevant for $d < 3$. If we instead use the $\mathcal{O}(\epsilon)$ result of the $d = 4 - \epsilon$ calculation of [43, 44] even for $\epsilon > 1$, we find that it becomes relevant for $d < 25/9$.

Similarly, using (III.65) to replace $\delta\rho$ in nonlinearities of type (iii), we find that they yield terms nonlinear in θ that involve *two* derivatives; we have ignored such terms from the start. Indeed, the coefficients of these nonlinearities scale either as $b^{z-1-\zeta+\chi}$ or as $b^{z-2+2\chi}$ and are less relevant than the families of nonlinearities in (III.20).

IV. IMMORTAL FLOCKS

A. Equation of motion

We turn now to the long-time, large-scale dynamics of the ordered phase of an *immortal* flock, which we remind the reader is a flock in which the number of flockers is conserved. In the absence of noise, this ordered state will simply be any one of the infinity of uniform steady states

(III.1) of the equations of motion (II.1) and (II.2)

$$\mathbf{p}(\mathbf{r}, t) = p_0 \hat{\mathbf{n}} \quad , \quad \rho(\mathbf{r}, t) = \rho_0 = \text{constant} \quad , \quad (\text{IV.1})$$

where p_0 is the value of the magnitude of the polarization at which $U(\rho_0, |\mathbf{p}|)$ vanishes.

Precisely as we did for Malthusian flocks, we now study fluctuations about this uniform state by writing the polarization as

$$\mathbf{p}(\mathbf{r}, t) = (p_0 + \delta p(\mathbf{r}, t))(\cos \theta(\mathbf{r}, t), \sin \theta(\mathbf{r}, t)) \quad , \quad (\text{IV.2})$$

and expand the density around its mean value ρ_0 :

$$\rho(\mathbf{r}, t) = \rho_0 + \delta \rho(\mathbf{r}, t) \quad . \quad (\text{IV.3})$$

Paralleling our discussion for Malthusian flocks, we expand (II.1) to first order in fluctuations $\delta p(\mathbf{r}, t)$ in the magnitude of the polarization $\mathbf{p}$, and $\delta \rho$ to obtain

$$(\cos \theta, \sin \theta) \partial_t \delta p + p_0 (-\sin \theta, \cos \theta) \partial_t \theta = -(\alpha \delta p + \bar{\alpha} \delta \rho)(\cos \theta, \sin \theta) + \dots \quad , \quad (\text{IV.4})$$

where we've defined $\alpha \equiv -p_0 \left(\frac{\partial U(\rho, |\mathbf{p}|)}{\partial |\mathbf{p}|} \right)_{\rho_0, p_0}$ and $\bar{\alpha} \equiv -p_0 \left(\frac{\partial U(\rho, |\mathbf{p}|)}{\partial \rho} \right)_{\rho_0, p_0}$ and the ellipsis denotes terms involving spatial derivatives of θ , δp , and $\delta \rho$ as well as products of δp , $\delta \rho$ and spatial derivatives of θ . Projecting this along $\hat{\mathbf{n}}$, here too we find that δp fluctuations are “fast”, and can therefore be eliminated by solving its equation of motion in the long-time limit, and inserting the result into the equations of motion for the fields $\theta(\mathbf{r}, t)$ and $\delta \rho(\mathbf{r}, t)$.

The result of this calculation is a closed set of equations of motion for those two fields, whose form is required by the symmetry of rotation invariance, and the conservation of particle number. We directly write these down without going through the algebra described in Sec. III A to obtain them from (II.1) and (II.2). These equations read:

$$\begin{aligned} \partial_t \rho = & -\nabla \cdot [(\beta_0 + w_1 \delta \rho + w_2 \delta \rho^2)(\cos \theta, \sin \theta)] \\ & + D_{\theta\rho} \partial_x \partial_y \theta + D_{x\rho} \partial_x^2 \delta \rho + D_{y\rho} \partial_y^2 \delta \rho + \nabla \cdot \boldsymbol{\xi}_\rho \quad , \quad (\text{IV.5}) \end{aligned}$$

$$\begin{aligned} \partial_t \theta = & (\lambda + g_1 \delta \rho) \nabla \cdot (-\sin \theta, \cos \theta) + D_x \partial_x^2 \theta + D_y \partial_y^2 \theta \\ & - (-\sin \theta, \cos \theta) \cdot \nabla (c_s \delta \rho + g_3 \delta \rho^2) + D_{\rho\theta} \partial_x \partial_y \delta \rho + \xi_\theta \quad , \quad (\text{IV.6}) \end{aligned}$$

where the Gaussian, zero-mean white noises have correlations

$$\begin{aligned} \langle \xi_\theta(\mathbf{x}, t) \xi_\theta(\mathbf{x}', t') \rangle &= 2\Delta_\theta \delta(\mathbf{x} - \mathbf{x}') \delta(t - t') \quad , \quad (\text{IV.7}) \\ \langle \xi_{\rho i}(\mathbf{x}, t) \xi_{\rho j}(\mathbf{x}', t') \rangle &= 2\Delta_\rho \delta_{ij} \delta(\mathbf{x} - \mathbf{x}') \delta(t - t') \quad (\text{IV.8}) \end{aligned}$$

In writing these equations, we have stopped the expansion in powers of $\delta \rho$ at $O(\delta \rho^2)$, in anticipation of finding a negative value of the density roughness exponent χ_ρ , which will make higher powers of $\delta \rho$ less relevant.

We can think of β_0 , w_1 , and w_2 as the expansion coefficients of β in equation (II.2).

These equations are invariant under the symmetry transformation in Eq. (III.5), by construction. Note that the dynamical equation for the Nambu-Goldstone mode in [16] is not symmetric under (III.5) and, therefore, it is incorrect. Further, arguments of Ref. [16] for asserting that various nonlinear terms are not renormalised are also incorrect. In fact, as we will discuss below, unlike in Malthusian flocks, we cannot find *any* hyperscaling relation in immortal flocks.

Note that the time derivative of θ as given by (IV.6) is *not* a total divergence. As we noted in section (III), and contrary to the claim in reference [13], there is no reason why it, or in general the rate of change of any Nambu-Goldstone mode, should be one. Hence, [13]’s starting equation of motion for the immortal case, and, therefore, the exponents they derive from it, are incorrect. Furthermore, comparison with (IV.6) suggests that their claimed numerical generation of a term $\propto \theta$, in the θ equation, by simulating Eq. (22), (23) of their SI (see Fig. S4 of [13]) is a result of inadvertently breaking *rotation* symmetry. Our analysis makes it clear that for Chaté and Solon’s [13] Eq. 18 to be rotation invariant, their $h_{x,2}$ has to be equal to λ_1 (or equivalently, $h_{x,2} = -\tilde{\sigma}$ in (22), (23) in their SI); see Eq. (III.5), and Eqs. (IV.5) and (IV.6) of this article. Indeed it is only for that case that their Fig. S4 shows a massless correlator. Fig. S4 shows that other choices of $h_{x,2}$ give a mass to their ϕ field, unsur-

prisingly, since they break rotation invariance regardless of the value of h_{x1} . As all nonlinear terms in Eqs. (IV.5) and (IV.6) are invariant under the transformation in Eq. (III.5) and a term $\propto \theta$ is not, no such term can be generated.

B. Linear theory

The linear theory of the immortal flock is discussed in detail in [11]. The key results of this analysis are as fol-

$$D_{\pm}(\phi) = \left(\frac{D_x + D_{x\rho}}{2} \right) \cos^2 \phi + \left(\frac{D_y + D_{y\rho}}{2} \right) \sin^2 \phi \pm \frac{1}{4} \left(\frac{\cos \phi}{c_{\pm}(\phi) + (w_1 + \lambda) \cos \phi} \right) \left[\left\{ (D_{y\rho} - D_y) \sin^2 \phi + (D_{x\rho} - D_x) \cos^2 \phi \right\} (w_1 - \lambda) + 2(\beta D_{\rho\theta} + c_s D_{\theta\rho}) \sin^2 \phi \right]. \quad (\text{IV.11})$$

In contrast to the Malthusian case, the wavevector dependence (IV.9) of the eigenfrequencies appears to leave some ambiguity as to the value of the dynamical exponent z : the linear dependence of the real ($c_{\pm}$) part suggests $z = 1$, while the quadratic form of the imaginary ($D_{\pm}$) part implies $z = 2$. However, since it is the imaginary part (i.e., the damping) that controls the scale of the angular fluctuations (as we'll see explicitly in a moment), the choice $z = 2$ is the correct one for the linear theory.

To see this, we begin with the spatio-temporal Fourier transform of the correlation functions $C^{\rho}(\mathbf{r}, t) \equiv \langle \rho(\mathbf{0}, 0) \rho(\mathbf{r}, t) \rangle$ and $C^{\theta}(\mathbf{r}, t) \equiv \langle \theta(\mathbf{0}, 0) \theta(\mathbf{r}, t) \rangle$ obtained from the linear theory:

$$C_{\mathbf{q}, \omega}^{\rho} = \frac{\beta^2 q^2 \sin^2 \phi \Delta_{\theta}}{[(\omega - c_- q)^2 + D_-^2 q^4][(\omega - c_+ q)^2 + D_+^2 q^4]} \quad (\text{IV.12})$$

$$C_{\mathbf{q}, \omega}^{\theta} = \frac{(\omega - w_1 q \cos \phi)^2 \Delta_{\theta}}{[(\omega - c_- q)^2 + D_-^2 q^4][(\omega - c_+ q)^2 + D_+^2 q^4]} \quad (\text{IV.13})$$

Integrating these overall frequency gives the spatially Fourier-transformed equal-time correlation functions:

$$C_{\mathbf{q}}^{\rho} = \frac{\beta^2 (D_+ + D_-) \sin^2 \phi \Delta_{\theta}}{2q^2 (c_+ - c_-)^2 D_+ D_-} \quad (\text{IV.14})$$

$$C_{\mathbf{q}}^{\theta} = \frac{[D_- (c_+ - w_1 \cos \phi)^2 + D_+ (c_- - w_1 \cos \phi)^2] \Delta_{\theta}}{2q^2 (c_+ - c_-)^2 D_+ D_-}. \quad (\text{IV.15})$$

From these, we can see that it is the imaginary (i.e., the damping) parts $D_{\pm}$ of the eigenfrequencies that control the scale of the fluctuations. To make this clear, note that the equal-time correlation functions only depend on

low: the eigenfrequencies $\omega_{\pm}(\mathbf{q})$ of modes with wavevector $\mathbf{q} = q(\cos \phi, \sin \phi)$, where ϕ is the angle between the wavevector $\mathbf{q}$ and the ordering direction $\hat{x}$, are given by

$$\omega_{\pm} = c_{\pm}(\phi)q - iD_{\pm}(\phi)q^2 \quad (\text{IV.9})$$

with the direction ϕ -dependent sound speeds $c_{\pm}(\phi)$ and dampings $D_{\pm}(\phi)$ given by

$$c_{\pm}(\phi) = \left(\frac{w_1 + \lambda}{2} \right) \cos \phi \pm \sqrt{\frac{(w_1 - \lambda)^2}{4} \cos^2 \phi + c_s \beta \sin^2 \phi}, \quad (\text{IV.10})$$

ratios of the speeds β , c_s , w_1 , and λ , not on the absolute values of those speeds themselves. Thus, the scale of the fluctuations remains fixed if we keep the diffusion constants D_x , D_y , $D_{x\rho}$, $D_{y\rho}$, $D_{\rho\theta}$, and $D_{\theta\rho}$ fixed. This implies that the linear theory will have the dynamical exponent $z = 2$.

Since the spatially Fourier transformed density $\rho(\mathbf{q}, t)$ and angle $\theta(\mathbf{q}, t)$ mean squared fluctuations both scale as $\frac{1}{q^2}$, the mean squared real space fluctuations of both, in the linear theory, will scale in two dimensions like $\ln(L)$, where L is the system size, just as the angle fluctuations do in the linear theory of Malthusian flocks studied in section (III). Hence, the “roughness” exponents χ and χ_{ρ} for the fields θ and $\delta\rho$, respectively, are $\chi = \chi_{\rho} = 0$ in the linear theory.

The scaling of angular fluctuations of both immortal and Malthusian flocks within a linear theory are equivalent to those of equilibrium polar or nematic liquid crystals. Unlike in equilibrium systems (away from a critical point), active currents make the density fluctuations as large as the Nambu-Goldstone fluctuations [55, 56]. These linear results are well-established. It is also well-known that they are modified by nonlinearities. We will re-examine the effect of nonlinearities on the scaling of hydrodynamic correlators via the standard renormalisation group procedure.

C. Dynamical Renormalization group for immortal flocks

The dynamical renormalization group analysis of the equations of motion (IV.5) and (IV.6) proceeds exactly as in the Malthusian case discussed in section III. The

only difference is that now we must consider the density

field as well. Our rescalings are:

$$y \rightarrow by, \quad t \rightarrow b^z t, \quad x \rightarrow b^\zeta x, \quad \theta \rightarrow b^\chi \theta, \quad \rho \rightarrow b^{\chi_\rho} \rho. \quad (\text{IV.16})$$

As with the Malthusian problem, we once again begin by expanding the equations of motion (IV.5) and (IV.6) for $\delta\rho$ and θ in powers of θ . This is easily seen to lead to

$$\begin{aligned} \partial_t \rho = & \sum_{n=1}^{\infty} k_n \partial_x (\theta^{2n}) + \sum_{n=0}^{\infty} \left(v_n \partial_y (\theta^{2n+1}) + d_n \partial_x (\theta^{2n} \delta\rho) + f_n \partial_y (\theta^{2n+1} \delta\rho) + s_n \partial_x (\theta^{2n} \delta\rho^2) + h_n \partial_y (\theta^{2n+1} \delta\rho^2) \right) \\ & + D_{\theta\rho} \partial_x \partial_y \theta + D_{x\rho} \partial_x^2 \delta\rho + D_{y\rho} \partial_y^2 \delta\rho + \nabla \cdot \xi_\rho \end{aligned} \quad (\text{IV.17})$$

$$\begin{aligned} \partial_t \theta = & \sum_{n=0}^{\infty} \left(a_n \theta^{2n} \partial_x \theta + c_n \theta^{2n+1} \partial_y \theta + \Upsilon_n \delta\rho \theta^{2n} \partial_x \theta + \Xi_n \delta\rho \theta^{2n+1} \partial_y \theta + \gamma_n \theta^{2n+1} \partial_x \delta\rho + e_n \theta^{2n} \partial_y \delta\rho \right. \\ & \left. + \mu_n \theta^{2n+1} \partial_x (\delta\rho^2) + \sigma_n \theta^{2n} \partial_y (\delta\rho^2) \right) + D_x \partial_x^2 \theta + D_y \partial_y^2 \theta + D_{\rho\theta} \partial_x \partial_y \delta\rho + \xi_\theta, \end{aligned} \quad (\text{IV.18})$$

where the expansion coefficients before we begin renormalizing (i.e., at DRG “time” $\ell = 0$), are given by

$$k_n(\ell = 0) = -\frac{(-1)^n \beta}{(2n)!}, \quad (\text{IV.19a})$$

$$v_n(\ell = 0) = -\frac{(-1)^n \beta}{(2n+1)!}, \quad (\text{IV.19b})$$

$$d_n(\ell = 0) = -\frac{(-1)^n w_1}{(2n)!}, \quad (\text{IV.19c})$$

$$f_n(\ell = 0) = -\frac{(-1)^n w_1}{(2n+1)!}, \quad (\text{IV.19d})$$

$$s_n(\ell = 0) = -\frac{(-1)^n w_2}{(2n)!}, \quad (\text{IV.19e})$$

$$h_n(\ell = 0) = -\frac{(-1)^n w_2}{(2n+1)!}, \quad (\text{IV.19f})$$

$$c_n(\ell = 0) = -\frac{(-1)^n \lambda}{(2n+1)!}, \quad (\text{IV.19g})$$

$$a_n(\ell = 0) = -\frac{(-1)^n \lambda}{(2n)!}, \quad (\text{IV.19h})$$

$$\Upsilon_n(\ell = 0) = -\frac{(-1)^n g_1}{(2n)!}, \quad (\text{IV.19i})$$

$$\Xi_n(\ell = 0) = -\frac{(-1)^n g_1}{(2n+1)!}, \quad (\text{IV.19j})$$

$$\gamma_n(\ell = 0) = \frac{(-1)^n c_s}{(2n+1)!}, \quad (\text{IV.19k})$$

$$e_n(\ell = 0) = -\frac{(-1)^n c_s}{(2n)!}, \quad (\text{IV.19l})$$

$$\mu_n(\ell = 0) = \frac{(-1)^n g_3}{(2n+1)!}, \quad (\text{IV.19m})$$

$$\sigma_n(\ell = 0) = -\frac{(-1)^n g_3}{(2n)!}. \quad (\text{IV.19n})$$

Note that the parameters v_0 , c_0 , and e_0 in (IV.18) and (IV.17) are precisely the parameters $-\beta$, $-\lambda$, and $-c_s$ appearing in the linear theory of section (IV B).

As with the Malthusian problem, these expressions will not continue to hold as we iterate the DRG. However, once again it is only the rescaling step of the DRG that changes the ratios $k_n/k_{n'}$, $v_n/v_{n'}$, and, most importantly, $v_n/k_{n'}$, because, before rescaling, these terms must resum to the $\nabla \cdot [(\beta(\cos\theta, \sin\theta))]$ term in (IV.17). Likewise, if one replaces k_n and v_n with the pairs (d_n, f_n) , (s_n, h_n) , (a_n, c_n) , (Υ_n, Ξ_n) , (γ_n, e_n) , and (μ_n, σ_n) in the previous sentence, it remains true; that is, those ratios remain fixed before rescaling as well.

The recursion relations between the parameters of (IV.17) and (IV.18) on successive steps of the RG are

$$k'_n = b^{2n\chi - \chi_\rho + z - \zeta} (k_n + \text{graphs}), \quad (\text{IV.20a})$$

$$v'_n = b^{(2n+1)\chi - \chi_\rho + z - -1} (v_n + \text{graphs}), \quad (\text{IV.20b})$$

$$d'_n = b^{2n\chi + z - \zeta} (d_n + \text{graphs}), \quad (\text{IV.20c})$$

$$f'_n = b^{(2n+1)\chi+z-1}(f_n + \text{graphs}), \quad (\text{IV.20d})$$

$$s'_n = b^{2n\chi+\chi_\rho+z-\zeta}(s_n + \text{graphs}), \quad (\text{IV.20e})$$

$$h'_n = b^{(2n+1)\chi+\chi_\rho+z-1}(h_n + \text{graphs}), \quad (\text{IV.20f})$$

$$a'_n = b^{2n\chi+z-\zeta}(a_n + \text{graphs}), \quad (\text{IV.20g})$$

$$c'_n = b^{(2n+1)\chi+z-1}(c_n + \text{graphs}), \quad (\text{IV.20h})$$

$$\Upsilon'_n = b^{2n\chi+\chi_\rho+z-\zeta}(\Upsilon_n + \text{graphs}), \quad (\text{IV.20i})$$

$$\Xi'_n = b^{(2n+1)\chi+\chi_\rho+z-1}(\Xi_n + \text{graphs}), \quad (\text{IV.20j})$$

$$\gamma'_n = b^{2n\chi+\chi_\rho+z-\zeta}(\gamma_n + \text{graphs}), \quad (\text{IV.20k})$$

$$e'_n = b^{(2n-1)\chi+\chi_\rho+z-1}(e_n + \text{graphs}), \quad (\text{IV.20l})$$

$$\mu'_n = b^{2n\chi+2\chi_\rho+z-\zeta}(\mu_n + \text{graphs}), \quad (\text{IV.20m})$$

$$\sigma'_n = b^{(2n-1)\chi+2\chi_\rho+z-1}(\sigma_n + \text{graphs}), \quad (\text{IV.20n})$$

$$D'_{\theta\rho} = b^{z+\chi-\chi_\rho-1-\zeta}(D_{\theta\rho} + \text{graphs}), \quad (\text{IV.20o})$$

$$D'_{x\rho} = b^{z-2\zeta}(D_{x\rho} + \text{graphs}), \quad (\text{IV.20p})$$

$$D'_{y\rho} = b^{z-2\zeta}(D_{y\rho} + \text{graphs}), \quad (\text{IV.20q})$$

$$D'_x = b^{z-2\zeta}(D_x + \text{graphs}), \quad (\text{IV.20r})$$

$$D'_y = b^{z-2\zeta}(D_y + \text{graphs}), \quad (\text{IV.20s})$$

$$D'_{\rho\theta} = b^{z+\chi_\rho-\chi-\zeta-1}(D_{\rho\theta} + \text{graphs}), \quad (\text{IV.20t})$$

$$\Delta'_\rho = b^{z-2\chi_\rho-\zeta-1}(\Delta_\rho + \text{graphs}), \quad (\text{IV.20u})$$

$$\Delta'_\theta = b^{z-2\chi-\zeta-1}(\Delta_\theta + \text{graphs}), \quad (\text{IV.20v})$$

where the primes denote parameters after the RG step in question, and “graphs” denote the renormalizations arising from averaging over the short-wavelength degrees of freedom.

Again as we did for the Malthusian problem, here too we can use these recursion relations, even without evaluating the graphical corrections, to determine which of the nonlinearities k_n , v_n , d_n , f_n , s_n , h_n , a_n , c_n , Υ_n , Ξ_n ,

γ_n , e_n , μ_n , and σ_n are “relevant” in the RG sense. We do this by once again noting that in the absence of those nonlinearities, the graphical corrections must vanish.

Hence, if we choose the rescaling exponents z , ζ , and χ in (IV.20o)-(IV.20v) to keep the linear damping terms D_x , D_y , $D_{\theta\rho}$, $D_{x\rho}$, $D_{y\rho}$, and $D_{\rho\theta}$, and the noise strengths Δ_θ and Δ_ρ fixed (so that the scale of the fluctuations remains the same upon renormalization), then we can assess the relevance or irrelevance of the nonlinearities k_n and v_n when those nonlinearities are initially small (so that the graphical corrections can be ignored in the early RG steps) by asking simply whether or not they grow upon renormalization if they start out being small.

Ignoring the graphical corrections, we see that equations (IV.20r), (IV.20s), (IV.20u), and (IV.20v) imply that, to keep the linear damping terms D_x , D_y , and the noise strengths Δ_θ and Δ_ρ fixed, we must choose the rescaling exponents z , ζ , χ_ρ , and χ so that the following four conditions are satisfied:

$$\begin{aligned} z - 2 &= 0 & , & & z - 2\zeta &= 0 & , & & z - \zeta - 2\chi - 1 &= 0 \\ z - \zeta - 2\chi_\rho - 1 &= 0, \end{aligned} \quad (\text{IV.21})$$

whose solutions are trivially found to

$$z = 2 \quad , \quad \zeta = 1 \quad , \quad \chi = \chi_\rho = 0, \quad (\text{IV.22})$$

which the alert reader will recognize as the exponents we found earlier by inspecting the correlation function predicted by the linear theory.

It is also straightforward to see that these exponents also keep the remaining four linear damping parameters $D_{\theta\rho}$, $D_{x\rho}$, $D_{y\rho}$, and $D_{\rho\theta}$ fixed.

Using these exponents in (IV.20a)-(IV.20n), we find that every single nonlinear term ν_n , where ν_n can stand for any one of the parameters k_n through σ_n in equations (IV.20a)-(IV.20n) obeys an identical recursion relation, which reads

$$\nu'_n = b(\nu_n + \text{graphs}), \quad (\text{IV.23})$$

We see from this immediately that *all* of the double septuplets of infinities of nonlinearities k_n through σ_n are *relevant* (that is, grow upon renormalization) if they are *initially small* (so that the “graphs” can be neglected). This proves that the linear results are *never* valid in a two-dimensional immortal flock, except in the very special case in which every single one of the aforementioned double septuplets of infinities of nonlinearities k_n through σ_n vanish, which will never occur in practice.

The long wavelength physics will therefore always be changed by the nonlinearities and must be controlled by an RG fixed point at which at least one of the nonlinearities ν_n is non-zero.

We’ll now argue that only a finite (although, alas, distressingly large) subset of the infinity of nonlinearities in (IV.17) and (IV.18) actually need to be considered.

The argument for this is basically the same as that used in our discussion in section (III) of Malthusian flocks and

relies on arguing that the roughness exponents χ and χ_ρ are both less than zero.

Unfortunately, for reasons that we'll discuss in more detail in a moment, we do not have as compelling an argument for the negativity of χ and χ_ρ as we had for the negativity of χ in Malthusian flocks. However, it remains true that, as we noted in our discussion of Malthusian flocks, that χ must be negative in order to have a long-range ordered state. And χ_ρ must be negative, as can be seen by the following *reductio ad absurdum* argument: The renormalization group implies that

$$P(\delta\rho, L) = L^{-\chi_\rho} f(\delta\rho/L^{\chi_\rho}), \quad (\text{IV.24})$$

where $\delta\rho$ is the departure of the space averaged value of ρ in a cubic "sample box" of side L from its mean value ρ_0 , and $P(\delta\rho, L)$ is the probability distribution of that mean density departure. Hence, the mean value $\langle\rho\rangle$ of ρ in that sample box is given by

$$\langle\rho\rangle = \rho_0 + \int_{-\rho_0}^{\infty} f(\delta\rho/L^{\chi_\rho}) \delta\rho \, d\delta\rho, \quad (\text{IV.25})$$

where $f(x)$ is a positive definite scaling function. Changing variables of integration to $x = \delta\rho/L^{\chi_\rho}$, and assuming (as we will now prove by contradiction is impossible) that $\chi_\rho > 0$, we obtain, for L very large,

$$\begin{aligned} \langle\rho\rangle &= \rho_0 + L^{\chi_\rho} \int_{-\rho_0/L^{\chi_\rho}}^{\infty} f(x) x dx \\ &\approx \rho_0 + L^{\chi_\rho} \int_0^{\infty} f(x) x dx, \end{aligned} \quad (\text{IV.26})$$

where the approximate equality becomes asymptotically exact in the limit $L \rightarrow \infty$.

The $\int_0^{\infty} f(x) x dx$ in the last expression is positive definite, since $x > 0$ throughout the range of integration, and $f(x)$ is positive definite, being a probability distribution.

Since in the limit of $L \rightarrow \infty$ the left-hand side is equal to ρ_0 by definition, the second term on the right-hand side must vanish in the same limit, which is in contradiction with the assumption $\chi_\rho > 0$. Hence, $\chi_\rho < 0$.

Furthermore, the fact that the graphical corrections to the diffusion constants $D_{x,y,(x\rho),(y\rho),(\theta\rho),(\rho\theta)}$ are probably all positive tends to reduce χ and χ_ρ from zero. Unfortunately, unlike the Malthusian case, here (as we'll show in a moment) the noise strengths Δ_θ and Δ_ρ are also renormalized, and it's not obvious *a priori* that the increase in the fluctuations caused by this enhanced noise doesn't overwhelm the suppression of the fluctuations caused by the enhancement of the damping.

However, given the extensive numerical evidence [8] that a long-range ordered state *does* exist in immortal flocks, it seems safe to conclude that χ and χ_ρ are both negative. Given this, it can be immediately seen from the recursion relations (IV.20a)-(IV.20n) that the most relevant nonlinear terms of the set of ν_n 's are those with the smallest value of n ; that is, k_1, v_1, d_1, a_1 , and e_1 , and the $n = 0$ term of all the other parameters $c_n, f_n, \dots, \gamma_n$,

σ_n . We have not included $v_0 = -\beta$, $a_0 = -\lambda$, $d_0 = -w_1$, and $e_0 = -c_s$ in this list, since these are linear terms.

So the minimal theory for two-dimensional immortal flocks does not have an infinite number of potentially relevant nonlinearities. However, its finite number is still quite large (fourteen)⁸

In practice, perturbative calculations of the graphical corrections are reliable if the nonlinearities are small at the fixed point. As in the Malthusian problem, we only expect this to be true near the critical dimension for this problem, which is well-known [2-4, 11] to be four. Hence, accurate calculation of the graphical corrections in two dimensions is not only formidable (because of the large number of relevant nonlinearities), but actually impossible, barring non-perturbative miracles.

Furthermore, the situation for immortal flocks is actually somewhat worse than that for Malthusian flocks, because in the immortal case, we cannot obtain *any* exact scaling relation between the exponents. This is because, as we've now pointed out many times, not all of the nonlinearities in the equation of motion (IV.18) for θ can be written as total derivatives. This rules out the possibility of using our argument from the Malthusian case that Δ_θ gets no graphical renormalization, which eliminates the corresponding exact exponent relation (III.31). Note that [13] assert that this relation *does* hold, because they erroneously require that the time derivative of θ must be a total divergence, because of their mistaken belief that all Nambu-Goldstone modes must have this property. In fact, the relation (III.31) need not hold, and almost certainly does not.

The nonlinearities in the $\delta\rho$ equation of motion (IV.17) are, of course, necessarily total divergences, since, in an immortal flock, the space integral of $\delta\rho$ is conserved. However, as the noise in (IV.17) is conserving, the fact that the nonlinearities are total divergences does not protect Δ_ρ from renormalization.

We can't even obtain exact scaling relations like (III.55) that we derived in the Malthusian case by using the fact that the graphical corrections to λ_1 and λ_2 are equal. The reason we can't use the analogous argument for the seven pairs of nonlinearities here that must, pairwise, have identical graphical corrections, is that we do not know that both members of any given pair are non-zero at the fixed point. For example, if we *did* know that, say, d_1 and f_0 were both non-zero at the fixed point, then equations (IV.20c) and (IV.20d), taken together with the observation that the graphical corrections to both are equal, would imply

$$2\chi + z - \zeta = \chi + z - 1, \quad (\text{IV.27})$$

⁸ In a theory of the coupled density and rotational Nambu-Goldstone mode in arbitrary dimensions, seven of these nonlinearities become relevant below $d = 4$ as shown in [11], therefore even a perturbative calculation of the exponents near $d = 4$ is prohibitively difficult.

which would immediately imply $\zeta = 1 + \chi$. Unfortunately, since we do *not* know that both d_1 and f_0 are non-zero at the fixed point, an alternative possibility is that $\zeta < 1 + \chi$, f_0 is irrelevant at the fixed point, and d_1 is finite there. The reverse possibility— $\zeta > 1 + \chi$, d_1 is irrelevant at the fixed point, and f_0 is finite there—is equally likely. A third possibility, namely that *both* f_0 and d_1 vanish at the fixed point, also exists.

Clearly, an exhaustive list of all these possibilities would be both *exhausting* and uninformative. The bottom line is that there is very little we can say about the scaling exponents z , ζ , χ , and χ_ρ for the immortal case, beyond the observation that they are *not* those predicted by the linear theory, and that they *are* universal. All claims to the contrary in the literature [2, 3, 13, 14, 16], including those by one of the current authors, are based on erroneous arguments, as we have argued above.

Therefore, the only analytical assertions that we *can* make with confidence are (i) the hydrodynamics of immortal flocks *must* be modified by nonlinearities and (ii) c_0 *cannot* be the *only* relevant nonlinearity in $d = 2$. If not for (ii), we could calculate the exponents exactly in $d = 2$: $\chi = \chi_\rho = -1/5$, $\zeta = 3/5$, $z = 6/5$. However, as shown in Sec. IIIC, this fixed point is unstable to the a_1 nonlinearity. Unfortunately, this only rules out two sets of exponents out of infinitely many possibilities.

V. SUMMARY AND OUTLOOK

In this work, we have comprehensively re-examined the hydrodynamic theory of flocks in two dimensions, for both “Malthusian” (non-number-conserving) and “immortal” (number-conserving) systems. For Malthusian flocks, we have obtained two exact scaling relations between the three scaling exponents that characterize the long-distance, long-time behaviour of these systems. We have argued that such flocks display long-range order in two dimensions.

We further demonstrate that for *immortal* flocks, there is no analytical argument that gives *any* exact scaling relations between the scaling exponents. This makes it im-

possible to demonstrate that such systems display long-range order in two dimensions. Therefore, the strongest evidence for the existence of long-range order in immortal flocks stems from extensive numerical studies [8].

While the “canonical” exponents [2, 3] $z = 2(d + 1)/5$, $\zeta = (d + 1)/5$, $\chi = (3 - 2d)/5$ are incorrect for flocks in $d = 2$, they do hold for *incompressible* flocks in $d \geq 3$ [57].

Given that we have shown that we cannot obtain the exponents of flocks in $d = 2$ analytically, is there any nonequilibrium flock whose hydrodynamic properties we *can* calculate analytically in $d = 2$? Yes: incompressible flocks in the presence of both annealed [58–60] and quenched disorder [61–63]. While the long-range order in the former is due to the incompressibility constraint and not due to motion, the long-range order in the latter *requires* motility. Similarly, we can exactly calculate the hydrodynamic properties of flocks at solid or fluid interfaces of bulk fluids [64–66].

ACKNOWLEDGMENTS

A.M. thanks Cesare Nardini for useful discussions and acknowledges the support of ANR through the grant PSAM. AM also acknowledges a TALENT fellowship awarded by CY Cergy Paris Université. A.M. and S.R. would like to thank the Isaac Newton Institute for Mathematical Sciences, Cambridge, for support and hospitality during the programmes “New Statistical Physics in Living Matter: non-equilibrium states under Adaptive Control” and “Anti-diffusive dynamics: from sub-cellular to astrophysical scales” (EPSRC grant no. EP/R014604/1), and SR additionally for a Rothschild Distinguished Visiting Fellowship. This research was supported in part by grant NSF PHY-2309135 to the Kavli Institute for Theoretical Physics. S.R. was supported in part by a J C Bose Fellowship of the ANRF, India. L.C. acknowledges support by the National Science Foundation of China (under Grant No. 12274452), and thanks the MPI-PKS, where the early stage of this work was performed, for their support.

* leiming@cumt.edu.cn

† patrick.jentsch@embl.de

‡ c.lee@imperial.ac.uk

§ nyomaitra07@gmail.com

¶ sriram@iisc.ac.in

** jjt@uoregon.edu

[1] T. Vicsek, A. Czirók, E. Ben-Jacob, I. Cohen, and O. Shochet, *Physical review letters* **75**, 1226 (1995).

[2] J. Toner and Y. Tu, *Physical review letters* **75**, 4326 (1995).

[3] J. Toner and Y. Tu, *Physical review E* **58**, 4828 (1998).

[4] J. Toner, *The Physics of Flocking: Birth, Death, and Flight in Active Matter* (Cambridge University Press,

2024).

[5] M. C. Marchetti, J.-F. Joanny, S. Ramaswamy, T. B. Liverpool, J. Prost, M. Rao, and R. A. Simha, *Reviews of modern physics* **85**, 1143 (2013).

[6] J. Toner, Y. Tu, and S. Ramaswamy, *Annals of Physics* **318**, 170 (2005).

[7] S. Ramaswamy, *Annu. Rev. Condens. Matter Phys.* **1**, 323 (2010).

[8] H. Chaté, *Annual Review of Condensed Matter Physics* **11**, 189 (2020).

[9] N. D. Mermin and H. Wagner, *Physical Review Letters* **17**, 1133 (1966).

[10] P. C. Hohenberg, *Physical Review* **158**, 383 (1967).

- [11] J. Toner, Physical Review E **86**, 031918 (2012).
- [12] J. Toner, Physical review letters **108**, 088102 (2012).
- [13] H. Chaté and A. Solon, Phys. Rev. Lett. **132**, 268302 (2024).
- [14] H. Ikeda, Physical Review Letters **133**, 258301 (2024).
- [15] B. Mahault, F. Ginelli, and H. Chaté, Physical review letters **123**, 218001 (2019).
- [16] A. Amoretti, D. K. Brattan, and L. Martinoia, Physical Review E **110**, 054108 (2024).
- [17] E. Bertin, M. Droz, and G. Grégoire, Physical Review E—Statistical, Nonlinear, and Soft Matter Physics **74**, 022101 (2006).
- [18] J.-B. Caussin, A. Solon, A. Peshkov, H. Chaté, T. Dauxois, J. Tailleur, V. Vitelli, and D. Bartolo, Physical review letters **112**, 148102 (2014).
- [19] A. P. Solon, H. Chaté, and J. Tailleur, Physical review letters **114**, 068101 (2015).
- [20] D. Martin, H. Chaté, C. Nardini, A. Solon, J. Tailleur, and F. Van Wijland, Physical Review Letters **126**, 148001 (2021).
- [21] D. Martin, G. Spera, H. Chaté, C. Duclut, C. Nardini, J. Tailleur, and F. van Wijland, Journal of Statistical Mechanics: Theory and Experiment **2024**, 084003 (2024).
- [22] M. Besse, H. Chaté, and A. Solon, Physical Review Letters **129**, 268003 (2022).
- [23] F. Jülicher, J. Prost, and J. Toner, Physical Review E **106**, 054607 (2022).
- [24] W. Kung, M. Cristina Marchetti, and K. Saunders, Physical Review E—Statistical, Nonlinear, and Soft Matter Physics **73**, 031708 (2006).
- [25] I. Low and A. V. Manohar, Physical review letters **88**, 101602 (2002).
- [26] M. Kardar, G. Parisi, and Y.-C. Zhang, Physical Review Letters **56**, 889 (1986).
- [27] I. S. Aranson, S. Scheidl, and V. M. Vinokur, Physical Review B **58**, 14541 (1998).
- [28] G. Pisegna, S. Saha, and R. Golestanian, Proceedings of the National Academy of Sciences **121**, e2407705121 (2024).
- [29] A. Haldar, A. Sarkar, S. Chatterjee, and A. Basu, Physical Review E **108**, L032101 (2023).
- [30] A. Haldar, A. Sarkar, S. Chatterjee, and A. Basu, Physical Review E **108**, 034114 (2023).
- [31] A. Maitra, M. Lenz, and R. Voituriez, Physical Review Letters **125**, 238005 (2020).
- [32] A. Maitra, Annual Review of Condensed Matter Physics **16** (2024).
- [33] M. Fruchart, R. Hanai, P. B. Littlewood, and V. Vitelli, Nature **592**, 363 (2021).
- [34] L. Sieberer, G. Wachtel, E. Altman, and S. Diehl, Physical Review B **94**, 104521 (2016).
- [35] G. Wachtel, L. Sieberer, S. Diehl, and E. Altman, Physical Review B **94**, 104520 (2016).
- [36] E. Altman, L. M. Sieberer, L. Chen, S. Diehl, and J. Toner, Physical Review X **5**, 011017 (2015).
- [37] R. Daviet, C. P. Zelle, A. Asadollahi, and S. Diehl, arXiv preprint arXiv:2412.09677 (2024).
- [38] L. M. Sieberer, M. Buchhold, J. Marino, and S. Diehl, arXiv preprint arXiv:2312.03073 (2023).
- [39] H. Stark and T. Lubensky, Physical review E **67**, 061709 (2003).
- [40] P. C. Martin, O. Parodi, and P. S. Pershan, Physical Review A **6**, 2401 (1972).
- [41] J. Prost and N. A. Clark, Hydrodynamic properties of two-dimensionally ordered liquid crystals, in *Proc Intl Conf on Liquid Crystals, Bangalore, December 1979*, edited by S. Chandrasekhar (Heyden, Philadelphia, 1980) pp. 53–58.
- [42] D. Forster, D. R. Nelson, and M. J. Stephen, Physical Review A **16**, 732 (1977).
- [43] L. Chen, C. F. Lee, and J. Toner, Physical Review E **102**, 022610 (2020).
- [44] L. Chen, C. F. Lee, J. Toner, *et al.*, Physical Review Letters **125**, 098003 (2020).
- [45] A. Maitra and S. Ramaswamy, Physical review letters **123**, 238001 (2019).
- [46] R. Lahiri and S. Ramaswamy, Physical review letters **79**, 1150 (1997).
- [47] L. Chen and J. Toner, Physical review letters **111**, 088701 (2013).
- [48] D. R. Nelson and R. A. Pelcovits, Physical Review B **16**, 2191 (1977).
- [49] S. Mishra, R. A. Simha, and S. Ramaswamy, Journal of Statistical Mechanics: Theory and Experiment **2010**, P02003 (2010).
- [50] S. Shankar, S. Ramaswamy, and M. C. Marchetti, Physical Review E **97**, 012707 (2018).
- [51] H. Ikeda, Physical Review Letters **134**, 129901 (2025).
- [52] T. Hwa and M. Kardar, Physical review letters **62**, 1813 (1989).
- [53] J. Toner, Physical Review E **110**, 064604 (2024).
- [54] D. R. Nelson, Physical Review B **11**, 3504 (1975).
- [55] S. Ramaswamy, R. A. Simha, and J. Toner, Europhysics Letters **62**, 196 (2003).
- [56] J. Toner, The Journal of chemical physics **150**, 154120 (2019).
- [57] L. Chen, C. F. Lee, and J. Toner, New Journal of Physics **20**, 113035 (2018).
- [58] L. Chen, C. F. Lee, and J. Toner, Nature communications **7**, 12215 (2016).
- [59] L. Chen, C. F. Lee, A. Maitra, and J. Toner, Physical Review E **109**, L012601 (2024).
- [60] A. Maitra, P. Srivastava, M. C. Marchetti, S. Ramaswamy, and M. Lenz, Physical Review Letters **124**, 028002 (2020).
- [61] A. Maitra, Physical Review E **101**, 012605 (2020).
- [62] L. Chen, C. F. Lee, A. Maitra, and J. Toner, Physical Review E **106**, 044608 (2022).
- [63] L. Chen, C. F. Lee, A. Maitra, and J. Toner, Physical Review Letters **129**, 188004 (2022).
- [64] N. Sarkar, A. Basu, and J. Toner, Physical Review Letters **127**, 268004 (2021).
- [65] N. Sarkar, A. Basu, and J. Toner, Physical Review E **104**, 064611 (2021).
- [66] A. Maitra, Nature Physics **19**, 733 (2023).